Release & test report

Build record, installation pipeline and remaining release gates.

Release status: prototype. Linux VST3 and native automated checks passed. No Mac AU/VST3 binary, DMG, Apple signature or notarization result has been produced in this session.

Executed coverage

Check	Recorded result
Linux Release build	PASS - pinned JUCE 8.0.6, x86_64 VST3 and native editor/processor.
DSP tests	PASS - three sample rates, silence reset, octave-shift spectrum, correlation and mono cancellation.
Processor/editor smoke	PASS - rates 44.1/48/96 kHz, block sizes 1/64/512, three modes, FX extremes, finite output, mono, state and MIDI release/sustain/panic. Reverb: bypass, wet-only pre-delay, 192 room/toggle combinations and recall.
UI and persistence	PASS - native creation/resize, formations, groups, conductor and singer dragging, mute, preset save/load, favourites, A/B and undo/redo.
pluginval 1.0.4	PASS - Linux VST3 strictness 5, seed 12345, GUI checks skipped. Native editor has separate offscreen tests.
Sanitizers	PASS for standalone DSP helpers with ASan/UBSan. LeakSanitizer was disabled due environment restrictions.
Packaging	Static shell/XML/YAML checks only. No Apple packaging command was executed.

Raw evidence, source and the actual editor screenshot are included in TestEvidence. These checks do not replace listening tests or prove compatibility with any named Mac DAW.

Performance & sound quality

Observed render cost

Release build, 48 kHz, 256-sample blocks, two seconds of a 220 Hz stereo guide, this Linux environment. Budget is wall-clock render time divided by audio duration, not an OS CPU measurement.

Layers	Before	After	Observed reduction
2	16.93%	1.30%	About 92%
12	16.65%	3.71%	About 78%
32	17.15%	8.56%	About 50%

What changed

Inactive voice slots stop processing after fade-out. Pitch targets are computed every 16 samples and interpolated. Major mix/FX/conductor changes, MIDI gates and delay time are smoothed. Delay reads use fractional interpolation.

What these results do not establish

Mac performance, real-time dropout limits, leak freedom, realistic human singers, formant preservation, correct pronunciation or parity with a commercial AI choir plugin have not been established.

Listening plan

Use a short dry vocal with sustained notes, pitch changes, words and unvoiced consonants. Compare bypass, Tight Doubles, Intimate and Wide at matched loudness. First judge pitch stability and word clarity with effects off; then add width and ambience. Check headphones, speakers and mono.

Only after the simple unison sound is convincing should the project add a trained choir model. That requires appropriate rights to voices and recordings, a compatible inference engine and independent performance tests.

Build the Mac installer

Option A: a dedicated GitHub repository

1. Create or choose a repository for I.R.Choir.
2. Upload the contents of the project folder at its root, including the hidden .github folder.
3. Push to main/master, or run Build and audit Mac DMG from Actions.
4. The Mac runner builds arm64 and x86_64 AU/VST3, runs CTest, packages the DMG, and invokes pluginval and auval.
5. Download the installer artifact only after a successful workflow. Test evidence is offered separately.

Option B: build on your Mac

Install CMake and Xcode command line tools, then double-click Build-I.R.Choir.command or run bash Scripts/build-mac.sh from the project root. Read errors if a tool, dependency or test is missing; do not treat a failed run as an installer.

Installer contents

The resulting DMG is designed to contain a branded Apple Installer .pkg, the three PDF guides, audit text, licenses and corresponding source archives. The package installs precompiled AU and VST3 bundles. It does not compile anything on the recipient Mac.

Signing is optional and separate

The default output is unsigned/unnotarized. For signed distribution, existing Developer ID Application and Installer certificates must be in the Mac keychain. Set IRCHOIR_APP_IDENTITY and IRCHOIR_INSTALLER_IDENTITY. An existing IRCHOIR_NOTARY_PROFILE enables submission, acceptance checking and DMG stapling.

> No Apple credentials are included or requested inside the project files. VST2 and AAX are not part of this build. The branded Apple Installer flow and DMG packaging still require actual Mac execution.

Release gates & audit findings

Gate	Current status
Native Mac build	Pending: universal arm64/x86_64 build and architecture checks.
Mac plugin validation	Pending: auval plus VST3 validation, host scan and routing.
Installer verification	Pending: mount DMG, inspect payload, install, rescan, uninstall and preserve presets on a fresh Mac.
DAW reliability	Pending: real sessions, automation, reopen/recall, long playback and dropouts.
Musical evaluation	Pending: real-vocal comparison, consonant clarity, pitch transitions, mono and loudness matching.
AI singer identities	Not implemented. Synthetic filters are not a replacement.
Warning cleanup	Some aggressive compiler conversion warnings remain; explicit cleanup is appropriate before production.
Distribution rights	Review JUCE and any future model/data licenses before distributing.

Material fixes made

Corrected invalid JUCE GUI color identifiers; removed dotted native bundle filenames that broke VST3 loading; prevented formation recall from resetting singer positions; corrected mono panning and output-gain order; fixed stage/meter overlap and stale screenshot output; isolated preset tests from real user files.

Source and licensing

Original source is offered under AGPL-3.0-or-later; the full text is included. JUCE 8.0.6 has its own AGPL/commercial licensing. The default DMG script includes corresponding source and license copies. No commercial choir samples or AI weights are included.

This is an engineering build report, not a certification of sound quality, licensing compliance or public-release readiness. The executable test logs and later Mac runner results take precedence over assumptions.